📖 Human-to-Logic Dictionary

A quick translation guide from everyday English requirements into formal logic and set theory.

Human Expression / Requirement	Concept	Logical / Set Translation
"A happens if and only if B happens"	Biconditional	$A \leftrightarrow B$
"A is a sub-category of B"	Subset	$A \subseteq B$
"All items / Every item must..."	For All	$\forall x$
"At least one item..."	Exists	$\exists x$
"Both A and B must be true"	AND	$A \wedge B$
"Combine groups A and B"	Union	$A \cup B$
"Either A or B (can be both)"	OR (Inclusive)	$A \vee B$
"Either A or B (strictly one, not both)"	XOR (Exclusive)	$A \oplus B$
"Everything except A"	Complement / NOT	A^c or $\neg A$
"Exactly one item..."	Unique Existential	$\exists! x$
"If A happens, then B must happen"	Implication	$A \to B$
"Is a member of / belongs to category A"	Element of	$x \in A$
"Is NOT a member of category A"	Not Element of	$x \notin A$
"Items in A, but exclude items in B"	Difference	$A - B$
"Neither A nor B"	NOR	$\neg(A \vee B)$ which is $\neg A \wedge \neg B$

Human Expression / Requirement	Concept	Logical / Set Translation
"Not A / The opposite of A"	NOT	$\neg A$
"Only items common to both A and B"	Intersection	$A \cap B$

💡 Complex Translation Examples

Sometimes humans combine these requirements. Here is how they break down:

- **"If a user is an Admin, they can Edit or Delete, but not both at the same time."**
  - $Admin \to (Edit \oplus Delete)$
- **"All items in the cart must be either Digital or Physical."**
  - $\forall x \in Cart : x \in Digital \lor x \in Physical$
- **"At least one user has Administrator rights."**
  - $\exists x : x \in Administrators$